Orders Analysis: A Descriptive Case Study

How a 368-row French retail export, full of inconsistent spellings and broken values, became a clean, chart-backed read on where order value actually comes from.

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Type: Descriptive analytics / data visualization case study

Executive Summary

This project takes a messy, real-world-style orders dataset — 368 rows spanning six product categories and seven French regions over thirteen months — and turns it into a trustworthy picture of where order value comes from. After cleaning, 351 orders remained, worth €153,169 in total. The headline finding is that price, not order quantity, is the dominant driver of order value, and that this holds fairly evenly across a product mix where no single category dominates. The dataset also carries real data-quality scars worth knowing about before anyone acts on it: roughly 3.7% of orders sit outside a normal price range, and a genuine calculation bug in the original cleaning logic was found and corrected during this project, changing the category revenue totals by a meaningful margin. What follows explains how the data was cleaned, what it shows, and what a reasonable next step looks like.

Business Context

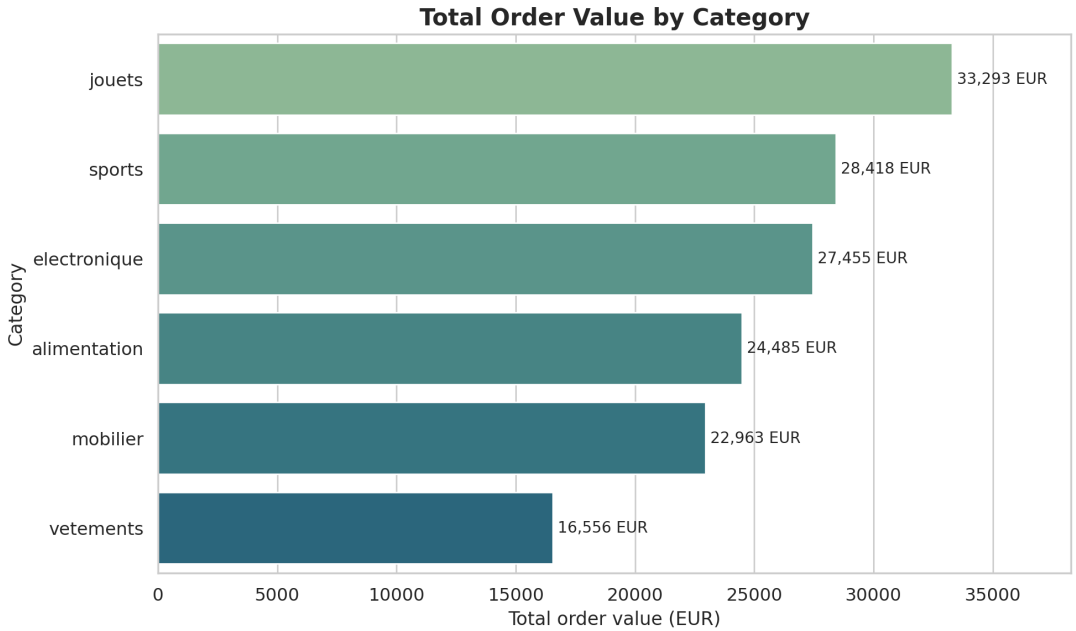
Picture a retail operations team that has just been handed a year's worth of order exports from a system that was clearly never designed with analysis in mind: region and category names typed by hand in at least three different conventions, prices recorded with stray minus signs and comma decimals, and a meaningful share of rows missing values entirely. Before anyone can answer a basic question — which category sells best, which region prices highest, whether order value is trending up or down — someone has to make the data trustworthy enough to answer those questions at all. That is the problem this project addresses: not building a predictive model, but establishing a clean, defensible baseline that a category manager or regional lead could actually use to make a call.

Methodology

The work followed four stages: cleaning, transformation, descriptive analysis, and visualization. Cleaning started with a full data-quality profile — missing-value rates ran as high as 27% in the raw order-amount column and 13% in order status — followed by removing 17 exact duplicate rows. Categorical fields (product, category, region, status) were normalized by stripping whitespace and accents and collapsing known abbreviations, since the same category could appear as "Électronique", "electro", or "ÉLECTRONIQUE" depending on who entered it. Prices and quantities were parsed out of inconsistent string formats (comma decimals, currency symbols, stray minus signs) into proper numeric types, and order value was recomputed from price times quantity rather than trusted from the dirty source column, which turned out to matter: the original fill-in logic filled order value independently of price and quantity, which silently broke that relationship for about 15% of rows once missing values were imputed. That bug was caught and fixed during this project, and it changes the category totals reported here versus a naive first pass. Once the data was consistent, descriptive analysis covered category and regional aggregation, IQR-based outlier detection on price, and a full correlation pass across the numeric fields, all of which fed directly into the five charts that follow.

Key Findings

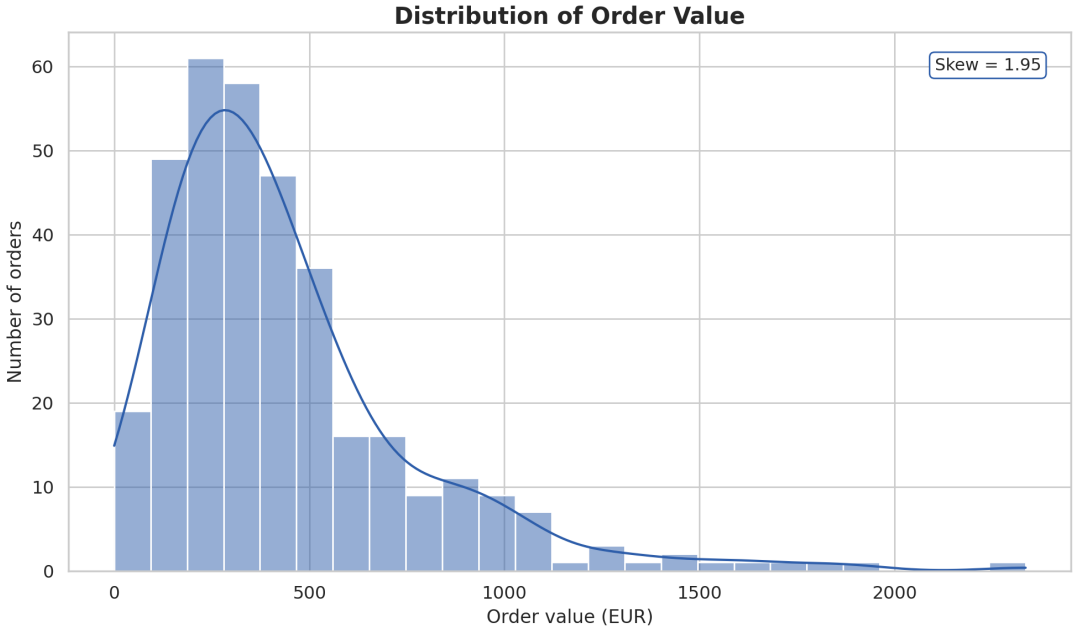
1. Category performance is broad-based, not top-heavy



Total order value by category, sorted descending.

Toys lead at €33,293 in total order value, but the gap to the lowest category, clothing at €16,556, is real without being dramatic — every category in between sits within a fairly narrow band. There is no single category carrying the business; a strategy built around "protect the top performer" would be missing five other categories doing comparable work.

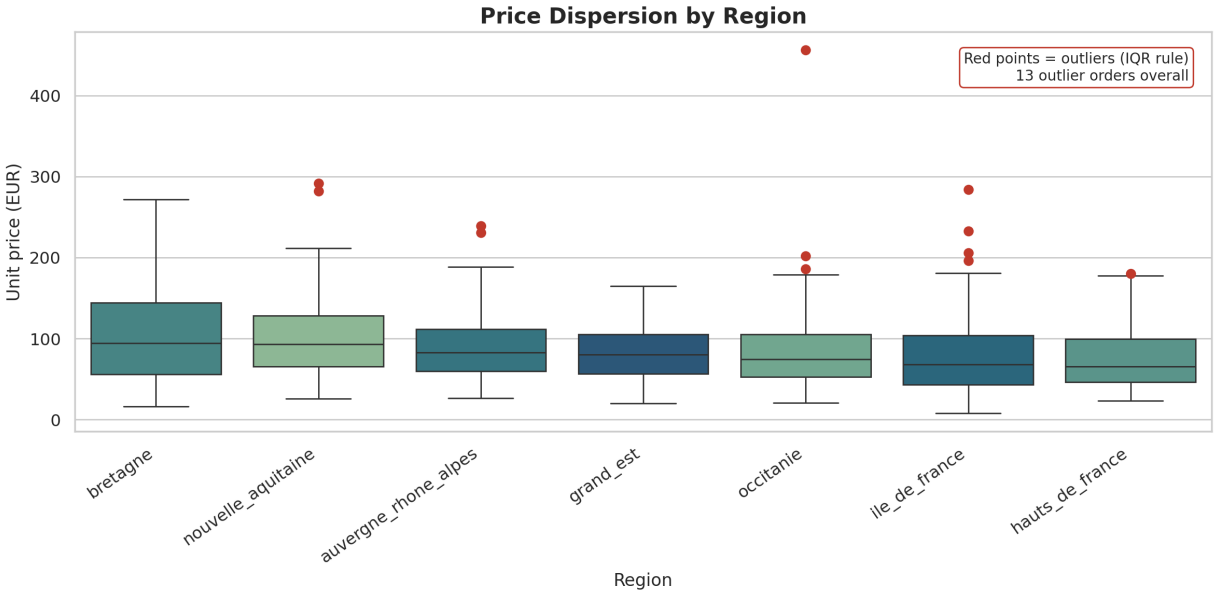
2. Most orders are modest, with a long tail of large ones



Distribution of order value across all cleaned orders.

The distribution is right-skewed, with a skew coefficient of 1.95. Most orders fall between €100 and €500, but a meaningful tail stretches out toward €2,300. Reporting a single average order value would flatter the typical customer's spend; the median (€354) is a far more honest read on what most orders actually look like.

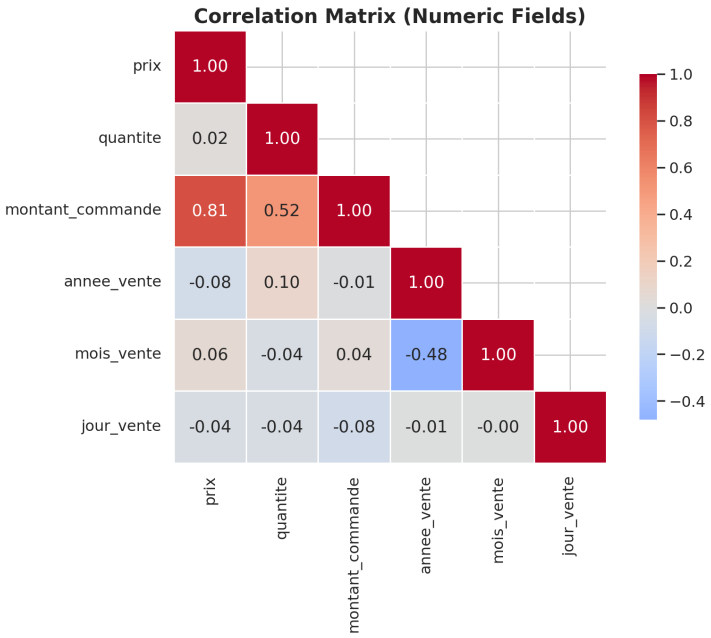
3. Regional pricing is not uniform



Unit price dispersion by region; red points mark IQR outliers.

Average unit price ranges from €106.60 in Bretagne down to €77.43 in Hauts-de-France — a spread of roughly 27%. Thirteen orders, 3.7% of the cleaned dataset, fall outside the normal IQR range on price, and they are not evenly spread: Nouvelle-Aquitaine and Île-de-France carry more than their share. That clustering is worth a manual look before anyone treats regional averages as directly comparable.

4. Price, not quantity, drives order value

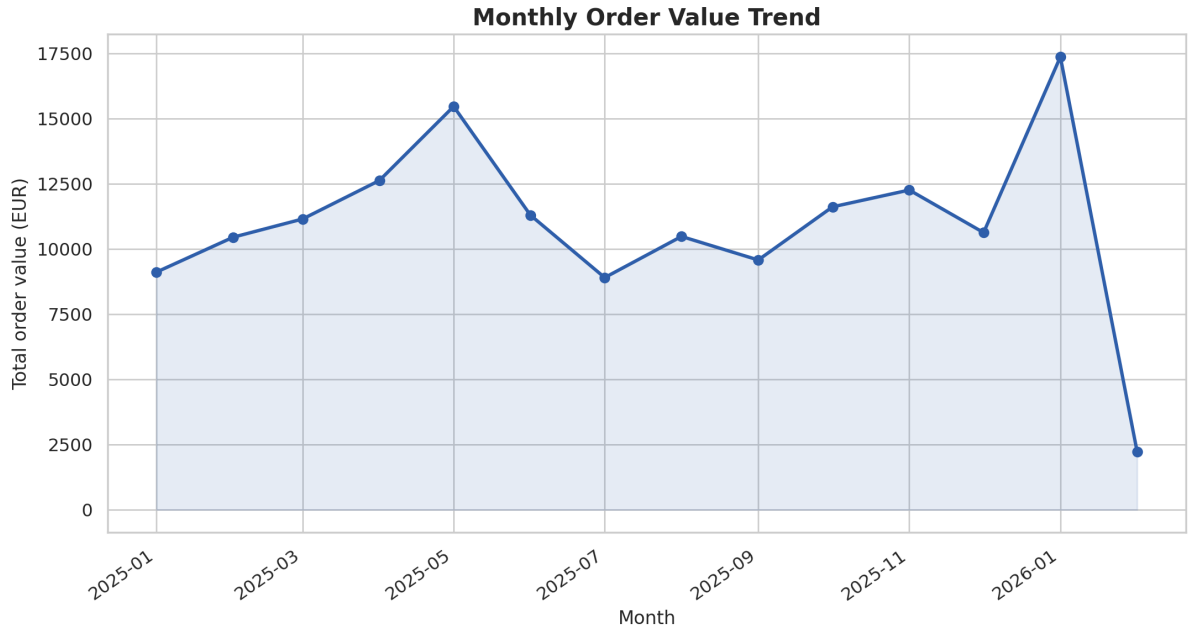


Correlation matrix across numeric order fields.

Unit price correlates with order value at r = 0.81, while quantity ordered correlates at a considerably weaker r = 0.52. In practical terms, orders get bigger mostly because items cost

more, not because customers buy more of them in one go. Any lever aimed at growing order value should start with pricing and product mix before it starts with basket size.

5. Revenue has two clear spikes against an otherwise steady baseline



Total order value by month (final month is partial).

Most months sit in a €9,000–€12,600 band, but May 2025 (€15,467) and January 2026 (€17,367) break well above it. February 2026 appears to fall off a cliff, but that month only has six orders on record in this export and should be read as incomplete data, not a real decline. The two genuine spikes are worth investigating against whatever promotional or seasonal calendar the business was running at the time.

Recommendations

First, treat pricing and product mix as the primary levers for order value, not basket size — the correlation gap between price and quantity is wide enough that a campaign built around "buy more items per order" is chasing the weaker signal. Second, run a manual audit of the thirteen flagged price outliers and the regional clustering behind them before using regional averages in any pricing decision; three or four bad rows in a dataset this size can meaningfully shift a regional mean. Third, before extending this analysis further, close the data-entry gaps at the source — the same category or region typed four different ways, and order amounts that don't match price times quantity, are cleanup costs that recur every time this export is pulled, and they are cheaper to fix upstream than downstream.

Technical Appendix

Built in Python 3.9+ with pandas and numpy for cleaning and aggregation, matplotlib and seaborn for all charts, and ReportLab for this document. The dataset totals 368 raw rows and 10 raw columns, reduced to 351 rows and 13 columns after deduplication and feature derivation (order value, and year/month/day extracted from the sale date). Known limitations: missing categorical and numeric values were filled using the column mode, which is a defensible default but not the

most precise option available; a group-aware fill (e.g. by product or category) would likely produce tighter estimates. A follow-up data-quality audit could still usefully investigate the thirteen flagged price outliers row-by-row, confirm whether the May 2025 and January 2026 spikes correspond to known promotional periods, and check whether the 133 orders marked "cancelled" or "returned" — 38% of the cleaned dataset — should be excluded from revenue totals rather than counted alongside completed orders, since the figures in this report include them all.

Closing

The most useful part of this project wasn't the charts — it was catching that the original cleaning logic quietly broke the price-times-quantity relationship for one in seven rows, and that a handful of price outliers cluster in specific regions rather than scattering randomly. Those are exactly the kind of issues that don't show up until someone actually sits with the data end to end. The next version of this project should push further into the data-quality side: source-level validation, a smarter imputation strategy, and a closer look at whether cancelled and returned orders belong in the same revenue figure as completed ones.